Stone–Jordan Exceptional Theory I: Exceptional-Algebra Structure behind the Standard-Model Skeleton—An Auditable Synthesis

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We assemble, in one auditable place, the chain of *known* algebraic facts that links normed division algebras, the exceptional Jordan (Albert) algebra $J_3(\mathbb{O})$, and the exceptional Lie groups F_4 and E_6 to the representation-theoretic "skeleton" of the Standard Model. We prove (or cite, with explicit attribution) only what is genuinely true: Hurwitz's classification of normed division algebras and the sedenion obstruction; the correspondence between 2×2 Hermitian matrices over $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ and Minkowski space in dimensions $3, 4, 6, 10$ (with the explicit correction that $H_2(\mathbb{H})$ is six-dimensional and that ordinary 4D Minkowski space is $H_2(\mathbb{C})$); the branchings $\mathbf{27} = \mathbf{1} + \mathbf{10} + \mathbf{16}$ and $\mathbf{16} = \mathbf{10} + \bar{\mathbf{5}} + \mathbf{1}$ together with the EIII coset $E_6/(\mathrm{Spin}(10) \times U(1))$ whose real tangent space is $\mathbf{16} \oplus \overline{\mathbf{16}}$; and the canonical tree-level value $\sin^2 \theta_W = 3/8$, stated with its renormalization-group caveat. None of these embeddings is new: each is standard since work of the 1930s–1980s, with modern syntheses by Baez, Baez–Huerta, Furey, Todorov–Dubois-Violette, Boyle, and Manogue–Dray–Wilson. The paper makes no mass fits, predicts no coupling, and does *not* derive the Standard Model. Its only contribution is organizational: a single "void/mirror" heuristic that motivates the chain, and a transparent ledger that cleanly separates the proven structural facts from a conjectural dynamical program (capacity selection of the algebraic partition, observer/family counting, a mirror-functor generation mechanism, and emergent dynamics), which we record as explicit open problems rather than results.

I. INTRODUCTION

A. A heuristic organizing motivation

The mathematics collected here was, for the author, motivated by a deliberately informal picture, which we label *heuristic* and use only as a narrative thread, never as an argument. Imagine a structureless "void" equipped with a single involutive operation—a "mirror" M—and ask what minimal algebraic worlds are forced once one demands that the mirror be compatible with a multiplicative norm. The mnemonic is only this: the normed/composition property survives exactly four doublings ($\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$) and then fails at the sedenions, and physics is imagined to live near that wall. We do *not* claim that iterating an involution derives or "selects" these algebras—Cayley–Dickson doubling does not preserve norm-composition (Prop. II.4, Rem. II.5), so the picture is a narrative thread about where the good structure stops existing, not a selection principle. The genuine selection question is left entirely open (Conj. VI.1); the slogan is suggestive and proves nothing.

We immediately flag two ways the slogan can mislead, and we correct both below. First, the naive expectation that a "mirror" obeys $M^2 = \mathrm{id}$ as an algebra symmetry is false beyond the octonions: in every Cayley–Dickson algebra conjugation *is* an involutive anti-automorphism with $x^{**} = x$, but the alternative law and the automorphism/triality structure that make this useful break at the sedenions (Sec. II). Second, the slogan must not be read as a derivation of dynamics; the entire dynamical content is relegated to the open-problem ledger of Sec. VI.

B. What this paper proves and what it leaves open

This paper proves nothing new. Every load-bearing statement below is either a classical theorem, restated with a short proof or a citation, or a standard branching rule from the unified-model-building literature. Concretely, we establish (by citation or short argument):

- the Hurwitz classification of normed division algebras as $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ and the failure of the normed/composition and alternative properties at the sedenions (Sec. II);
- the identification of $H_2(\mathbb{A})$ with $(n+2)$-dimensional Minkowski space for $n = \dim_{\mathbb{R}} \mathbb{A} \in \{1, 2, 4, 8\}$, giving dimensions $3, 4, 6, 10$, with the explicit correction that the quater-

nionic case is *six*-dimensional and that ordinary 4D spacetime is the *complex* case (Sec. III);

- the E_6 branchings $\mathbf{27} = \mathbf{1} + \mathbf{10} + \mathbf{16}$, the EIII coset decomposition $\mathbf{32} = \mathbf{16} \oplus \overline{\mathbf{16}}$, the grand-unification chain $E_6 \to \mathrm{SO}(10) \to \mathrm{SU}(5) \to \mathrm{SM}$, the identification of one generation with the $\mathbf{16}$, and the Spin(10) tensor products including $\mathbf{16} \times \mathbf{16} = \mathbf{10} + \mathbf{120} + \mathbf{126}$ (Sec. IV);

- the tree-level value $\sin^2\theta_W = 3/8$ in canonical GUT normalization, together with the honest statement that this is a high-scale boundary condition, not the measured number, and that the one-loop coefficient b_0 is content-dependent (Sec. V).

What we leave open is everything dynamical. We do not derive which algebra "wins," why there are three generations, how a mirror operation could generate matter, or what the equations of motion are. These are collected as explicit conjectures and open problems in Sec. VI and are *not* claimed as results.

C. Honest relation to prior work

The division-algebra and exceptional-group approach to particle physics is old and well developed, and we claim no priority over any of it. The octonionic origin of SU(3) color via $G_2 = \mathrm{Aut}(\mathbb{O})$ is due to Günaydin and Gürsey [26]. E_6 grand unification with a generation in the $\mathbf{27}$ is due to Gürsey, Ramond and Sikivie [27]. The group-theoretic identity $\sin^2\theta_W = 3/8$ and its renormalization-group running originate with Georgi–Glashow [21] and Georgi–Quinn–Weinberg [22]. The spacetime correspondence $H_2(\mathbb{A}) \cong \mathbb{R}^{1,n+1}$ dates to Kugo–Townsend [15] and Sudbery [14], streamlined by Baez–Huerta [13]. The branching tables are catalogued by Slansky [18], and the entire "Standard Model as representation theory" viewpoint is presented cleanly by Baez [1] and Baez–Huerta [28]. More recent programs include the complex-octonion Cl(6) constructions of Furey [29, 30] and Stoica [31]; the exceptional Jordan algebra constructions of Todorov–Dubois-Violette [32] and Boyle [33]; and the explicit octonionic E_8 organization of Manogue–Dray–Wilson [35]. For contrast we note the E_8 "theory of everything" proposals of Lisi [36, 37] and the Distler–Garibaldi no-go theorem [38] that constrains them. Because all four embeddings used here and the value

3/8 are pre-existing and standard, the present paper is a synthesis with an explicit honesty ledger, not a source of new physics or new mathematics.

II. DIVISION ALGEBRAS, THE ALBERT ALGEBRA $J_3(\mathbb{O})$, AND THE EXCEPTIONAL SERIES

A. Normed division algebras and the Hurwitz wall

Definition II.1. A *composition algebra* (normed division algebra) over $\mathbb{R}$ is a unital $\mathbb{R}$-algebra $\mathbb{A}$ equipped with a nondegenerate quadratic form N satisfying $N(xy) = N(x)N(y)$ for all $x, y \in \mathbb{A}$.

Theorem II.2 (Hurwitz)**.** *The only finite-dimensional real composition algebras with identity are* $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$*, of real dimensions* $1, 2, 4, 8$*. Here* $\mathbb{R}$ *and* $\mathbb{C}$ *are commutative and associative,* $\mathbb{H}$ *is associative but noncommutative, and* $\mathbb{O}$ *is neither commutative nor associative but is alternative (any subalgebra generated by two elements is associative).*

Proof. This is Hurwitz's composition-algebra theorem [2]; the modern proof shows that multiplicativity of the norm forces $\dim_{\mathbb{R}} \mathbb{A} \in \{1, 2, 4, 8\}$. See Baez [1, Sec. 2–3], Conway–Smith [9], or Springer–Veldkamp [7]. □

Remark II.3. We are careful with terminology. Hurwitz's theorem concerns *normed* (composition) division algebras with identity. The statement that any finite-dimensional real division algebra (no zero divisors, not necessarily normed or unital) has dimension $1, 2, 4$, or 8 is a separate, topological theorem of Bott–Milnor–Kervaire; it does not force isomorphism with $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$. We use only the Hurwitz statement.

B. Cayley–Dickson doubling and the sedenion obstruction

Proposition II.4 (Cayley–Dickson loss of properties)**.** *The Cayley–Dickson construction doubles an algebra* A *with conjugation to* A^2 *via* $(a, b)(c, d) = (ac - d^*b,\ da + bc^*)$ *and* $(a, b)^* = (a^*, -b)$*. Iterating from* $\mathbb{R}$ *gives* $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S}$ *(sedenions,* $\dim 16$*). At each step a property is lost:* $\mathbb{C}$ *loses order,* $\mathbb{H}$ *loses commutativity,* $\mathbb{O}$ *loses associativity (retaining alternativity), and* $\mathbb{S}$ *loses the alternative law and the normed-composition/division property:*

the sedenions contain zero divisors, e.g.

$$(e_1 + e_{10})(e_5 + e_{14}) = 0, \tag{1}$$

so $\mathbb{S}$ *is not a division algebra and* $N(xy) = N(x)N(y)$ *fails.*

Proof. The construction and its properties are standard (Schafer [6]; Baez [1, Sec. 2.2]). Existence of sedenion zero divisors is due to Moreno [11] and Cawagas [12]. □

Remark II.5 (The naive "mirror $M^2 = \mathrm{id}$" fails). In *every* Cayley–Dickson algebra, including $\mathbb{S}$ and beyond, conjugation $x \mapsto x^*$ is an involutive anti-automorphism: $x^{**} = x$ and $(xy)^* = y^*x^*$. Thus the conjugation map itself always squares to the identity. The heuristic "mirror" picture of Sec. I can therefore be salvaged only if M means precisely this conjugation. It *fails* if M is read as an order-two algebra automorphism tied to multiplication: once alternativity is lost, the left/right multiplication operators L_x, R_x need not square to scalars (in $\mathbb{S}$, L_x is not even injective for zero divisors), and the bimodule/triality symmetries of $\mathbb{O}$ break. The correct statement is that conjugation is always an order-two involution; what breaks at dimension 16 is the alternative law and the attendant norm/automorphism structure, not the involutivity of conjugation.

This is the precise sense in which "physics lives at a wall": the normed-composition chain terminates at $\mathbb{O}$ (dim 8), and the next double is already pathological.

C. The Albert algebra and the exceptional groups

Definition II.6. Let $h_3(\mathbb{O}) = J_3(\mathbb{O})$ denote the set of 3×3 octonion-Hermitian matrices ($X = X^*$, $*$ the conjugate-transpose) with the Jordan product $X \circ Y = \frac{1}{2}(XY + YX)$.

Theorem II.7 (Albert; Jordan–von Neumann–Wigner)**.** $J_3(\mathbb{O})$ *is a formally real, simple, finite-dimensional Jordan algebra of real dimension* 27*. It is the unique* exceptional *simple Jordan algebra, i.e. not isomorphic to a Jordan algebra of associative matrices under the anticommutator, precisely because* $\mathbb{O}$ *is nonassociative.*

Proof. The dimension count is 3 (real diagonal entries) + 3×8 (octonionic off-diagonal entries) $= 3 + 24 = 27$. Exceptionality and the classification are due to Jordan–von Neumann–Wigner [3] and Albert [4]; see Baez [1, Sec. 3] and Springer–Veldkamp [7, Ch. 5]. □

Remark II.8. We emphasize $\dim J_3(\mathbb{O}) = 27$, not 26 or 24. The trace-zero subspace $J_3(\mathbb{O})_0$ has dimension 26; it is a distinct object: the **26** of F_4 (the trace-zero representation) and the $\mathfrak{f}_4$-module coset in $\mathfrak{e}_6 = \mathfrak{f}_4 \oplus J_3(\mathbb{O})_0$. The minimal nontrivial irreducible representation of E_6 is the **27**, which is irreducible under E_6: the reduced structure group preserves the cubic determinant but *not* the linear trace, so the 26-dimensional trace-zero subspace is not E_6-invariant and is not an E_6 representation.

Theorem II.9 (Chevalley–Schafer)**.** *The automorphism group of $J_3(\mathbb{O})$ (linear maps preserving the Jordan product) is the compact, connected, simply connected exceptional Lie group F_4, of dimension* 52*; equivalently* $\operatorname{der} J_3(\mathbb{O}) = \mathfrak{f}_4$*. F_4 fixes the identity and acts transitively on rank-one idempotents with stabilizer* $\mathrm{Spin}(9)$*, so* $\mathbb{OP}^2 = F_4/\mathrm{Spin}(9)$*. The cubic determinant (Freudenthal) form on $J_3(\mathbb{O})$ is preserved by the real exceptional group of type E_6, of dimension* 78 *(the reduced structure group); F_4 is the subgroup of the structure group also fixing the identity, and Lie-algebraically $\mathfrak{e}_6 = \mathfrak{f}_4 \oplus J_3(\mathbb{O})_0$ with* $52 + 26 = 78$.

Proof. Chevalley–Schafer [5]; Freudenthal; Jacobson [8]; Baez [1, Sec. 4.2, 4.4]; Springer–Veldkamp [7, Ch. 7]. □

Remark II.10. $\mathrm{Aut}(J_3(\mathbb{O})) = F_4$ (dimension 52) is the Jordan-product-preserving group; the determinant-preserving structure group is E_6 (dimension 78). These must not be confused. The dimensions of the compact exceptional groups are $G_2 = 14$, $F_4 = 52$, $E_6 = 78$, $E_7 = 133$, $E_8 = 248$, obtained as $\dim = \mathrm{rank} + \#\mathrm{roots}$ (e.g. E_8: $8 + 240 = 248$); the common slip of quoting 240 (the root count) as $\dim E_8$ is to be avoided [10].

III. SPACETIME FROM 2×2 HERMITIAN MATRICES OVER DIVISION ALGEBRAS

Theorem III.1 (Minkowski space from $H_2(\mathbb{A})$)**.** *Let $\mathbb{A} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$ with $n = \dim_{\mathbb{R}} \mathbb{A}$. The space $H_2(\mathbb{A})$ of 2×2 Hermitian matrices over $\mathbb{A}$ is a real vector space of dimension $n + 2$, and the determinant*

$$\det \begin{pmatrix} t + x & y \\ y^* & t - x \end{pmatrix} = t^2 - x^2 - |y|^2, \qquad y \in \mathbb{A}, \tag{2}$$

is a nondegenerate quadratic form of Lorentzian signature. Hence $H_2(\mathbb{A}) \cong \mathbb{R}^{1,n+1}$, Minkowski space of dimension $n + 2$.

Proof. A Hermitian 2×2 matrix has the displayed form with $t, x \in \mathbb{R}$ and $y \in \mathbb{A}$, giving $1+1+n = n+2$ real parameters. The determinant $t^2 - x^2 - y^* y = t^2 - x^2 - |y|^2$ has one positive and $n+1$ negative directions, i.e. signature $(1, n+1)$. The formula is unambiguous even when $\mathbb{A}$ is noncommutative or nonassociative. See Baez–Huerta [13, Sec. 3] and Sudbery [14]. □

$\mathbb{A}$	$n = \dim_{\mathbb{R}} \mathbb{A}$	$\dim H_2(\mathbb{A}) = n+2$	signature	spacetime
$\mathbb{R}$	1	3	$(1,2)$	$\mathbb{R}^{1,2}$ (3D)
$\mathbb{C}$	2	4	$(1,3)$	$\mathbb{R}^{1,3}$ (ordinary 4D)
$\mathbb{H}$	4	6	$(1,5)$	$\mathbb{R}^{1,5}$ (6D)
$\mathbb{O}$	8	10	$(1,9)$	$\mathbb{R}^{1,9}$ (10D)

TABLE I. The corrected, standard correspondence between $H_2(\mathbb{A})$ and Minkowski space. Note that $H_2(\mathbb{H})$ is *six*-dimensional and that ordinary 4D Minkowski space is the *complex* block $H_2(\mathbb{C})$.

Remark III.2 (Two corrections we insist on). First, $H_2(\mathbb{H})$ is six-dimensional, with signature $(1, 5)$: the off-diagonal quaternionic entry contributes $\dim_{\mathbb{R}} \mathbb{H} = 4$ real directions, so $\dim = 1 + 1 + 4 = 6$. It is *not* four-dimensional, and the quaternionic case is not the case of ordinary spacetime. Second, ordinary 4D Minkowski space $\mathbb{R}^{1,3}$ is the *complex* case $H_2(\mathbb{C})$ ($\dim = 1 + 1 + 2 = 4$), with $\mathrm{SL}(2, \mathbb{C}) = \mathrm{Spin}(1, 3)$. Throughout, signatures are written as (timelike, spacelike)$= (1, n+1)$, and $\mathrm{Spin}(1, n+1)$ and $\mathrm{Spin}(n+1, 1)$ denote the same group (the choice is conventional).

Proposition III.3 (Spin groups)**.** *Determinant-preserving* $\mathbb{A}$*-linear transformations of* $H_2(\mathbb{A})$ *realize the double cover of the Lorentz group on the* $(n + 2)$*-dimensional vector representation:*

$$\mathrm{SL}(2, \mathbb{R}) = \mathrm{Spin}(1, 2), \quad \mathrm{SL}(2, \mathbb{C}) = \mathrm{Spin}(1, 3), \quad \mathrm{SL}(2, \mathbb{H}) = \mathrm{Spin}(1, 5). \tag{3}$$

For $\mathbb{A} = \mathbb{O}$ *the symbol "*$\mathrm{SL}(2, \mathbb{O})$*" is heuristic only—it is not a Lie group, since* $\mathbb{O}$ *is nonassociative (see Dray–Manogue [34] for the octonionic* $\mathrm{SL}(2, \mathbb{O})/E_6$ *spinor formalism)—but* $\mathrm{Spin}(9, 1)$ *nonetheless acts on the* 10*-dimensional vector representation* $H_2(\mathbb{O})$ *and on* 16*-dimensional spinor representations.*

Proof. $\mathrm{SL}(2, \mathbb{A})$ preserves the determinant (the Minkowski metric) and acts on the $n + 2$ vector representation, identifying it with the vector representation of $\mathrm{Spin}(n + 1, 1)$ [13,

p. 7]. The octonionic caveat (nonassociativity, and $\dim \mathrm{SO}(9,1) = 45$ versus only 32 real components of a 2×2 octonionic matrix) is discussed by Sudbery [14] and Baez [1]. $\square$

Remark III.4. Spinors in this framework are elements of $\mathbb{A}^2$, of real dimension $2n$; the pairing $\mathbb{A}^2 \times \mathbb{A}^2 \to H_2(\mathbb{A})$ reproduces the Clifford/γ-matrix structure. The vanishing of the relevant trilinear (the "three-psi" identity) holds exactly in dimensions $3, 4, 6, 10$, which is why classical super-Yang–Mills and the Green–Schwarz superstring are supersymmetric precisely there [13, 15, 16]. Note that the vector representation $H_2(\mathbb{A})$ (dimension $n + 2$) and the spinor representation $\mathbb{A}^2$ (real dimension $2n$) are distinct; for $\mathbb{O}$ they are 10 and 16 respectively. Octonionic Hermitian matrices can fail to have real eigenvalues [17], but the determinant formula is unaffected.

IV. THE E_6 FUNDAMENTAL AND THE STANDARD-MODEL SKELETON

A. The 27 and the EIII coset

Proposition IV.1 (Branching of the **27**)**.** *Under* $E_6 \supset \mathrm{Spin}(10) \times U(1)$*,*

$$\mathbf{27} = \mathbf{1}_{(+4)} + \mathbf{10}_{(-2)} + \mathbf{16}_{(+1)}, \tag{4}$$

in Slansky's integer normalization, and the adjoint splits as

$$\mathbf{78} = \mathbf{45}_{(0)} + \mathbf{1}_{(0)} + \mathbf{16}_{(-3)} + \overline{\mathbf{16}}_{(+3)}, \tag{5}$$

i.e. $\mathfrak{e}_6 = \mathfrak{so}(10) \oplus \mathbb{R} \oplus \mathbf{16} \oplus \overline{\mathbf{16}}$*. The charges are traceless:* $1 \cdot 4 + 10 \cdot (-2) + 16 \cdot 1 = 0$.

Proof. Standard branching tables of Slansky [18]; dimension checks $1 + 10 + 16 = 27$ and $45 + 1 + 16 + 16 = 78$. The $\mathbf{16}, \overline{\mathbf{16}}$ are the Weyl spinor of $\mathrm{Spin}(10)$ and its conjugate. $\square$

Proposition IV.2 (The EIII coset)**.** *The exceptional Hermitian symmetric space* $\mathrm{EIII} = E_{6(-14)}/(\mathrm{Spin}(10) \cdot U(1))$ *has real dimension* $78 - (45 + 1) = 32$ *and complex dimension* 16*, rank* 2*. Its isotropy (tangent) representation, as a real representation, is*

$$\mathbf{32} = \mathbf{16} \oplus \overline{\mathbf{16}}, \tag{6}$$

the complex $\mathrm{Spin}(10)$ *spinor plus its conjugate, with holonomy* $\mathrm{Spin}(10) \cdot U(1) \subset U(16)$.

Proof. This follows from the symmetric decomposition $\mathfrak{e}_6 = \mathfrak{k} \oplus \mathfrak{m}$ with $\mathfrak{k} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$ and $\mathfrak{m} = \mathbf{16} \oplus \overline{\mathbf{16}}$, the same off-diagonal blocks as in Proposition IV.1. See Helgason [19]. □

Remark IV.3 (No "28 + 4" branching). The honest representation-theoretic statement is $\mathbf{32} = \mathbf{16} \oplus \overline{\mathbf{16}}$ under $\mathrm{Spin}(10) \times U(1)$. Any split such as "28 + 4" (e.g. a choice of four internal plus twenty-eight directions) is an *extra* physical or geometric assumption, not a $\mathrm{Spin}(10) \times U(1)$ branching, and we do not present it as one.

B. The grand-unification chain and one generation

Proposition IV.4 (Breaking chain and one generation). $E_6 \supset \mathrm{SO}(10) \times U(1) \supset \mathrm{SU}(5) \times U(1) \times U(1) \supset \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times U(1)_Y$ *is the canonical maximal-subgroup chain of chiral grand unification, each step a regular embedding. Under* $\mathrm{SU}(5)$*,*

$$\mathbf{16} = \mathbf{10} + \bar{\mathbf{5}} + \mathbf{1}, \tag{7}$$

where the $\mathbf{10}$ *holds* (Q, u^c, e^c)*, the* $\bar{\mathbf{5}}$ *holds* (d^c, L)*, and the singlet is the right-handed neutrino* ν_R*; this is one anomaly-free generation. The full* $\mathbf{27} = \mathbf{16}_{(+1)} + \mathbf{10}_{(-2)} + \mathbf{1}_{(+4)}$ *contains, beyond this generation, only vectorlike exotics (the extra* $\mathbf{10} + \mathbf{1}$*), not a fourth chiral family; such* E_6 *exotics and their associated* Z' *phenomenology are reviewed in [20].*

Proof. Branching tables of Slansky [18]; the SU(5) step is Georgi–Glashow [21]; the synthesis is in Baez–Huerta [28]. □

Proposition IV.5 (Spin(10) spinor products). *For* Spin(10)*,*

$$\mathbf{16} \times \mathbf{16} = \mathbf{10}_s + \mathbf{120}_a + \mathbf{126}_s, \qquad \mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}, \tag{8}$$

with $\mathbf{10}$ *and* $\mathbf{126}$ *symmetric and* $\mathbf{120}$ *antisymmetric; both sides total* $256 = 16^2$*.*

Proof. Slansky [18, Table 42]. The dimensions are $\mathbf{45} = \binom{10}{2}$ (adjoint), $\mathbf{120} = \binom{10}{3}$, $\mathbf{210} = \binom{10}{4}$, and $\mathbf{126} = \frac{1}{2}\binom{10}{5}$ (a complex self-dual rank-five form). Note $\mathbf{16} \times \mathbf{16}$ contains no Spin(10) singlet, whereas $\mathbf{16} \times \overline{\mathbf{16}}$ does. □

Remark IV.6. The historical origin of E_6 unification with a generation in the $\mathbf{27}$ is Gürsey–Ramond–Sikivie [27]; E_6 has complex (non-self-conjugate) representations, so $\mathbf{27}$ and $\overline{\mathbf{27}}$ are inequivalent, with $\overline{\mathbf{27}} = \mathbf{1}_{(-4)} + \mathbf{10}_{(+2)} + \overline{\mathbf{16}}_{(-1)}$.

V. THE UNIFICATION OBSERVABLE $\sin^2\theta_W = 3/8$

Proposition V.1 (Trace ratio in canonical GUT normalization)**.** *Given the canonical* SU(5) *hypercharge normalization $T_Y = \sqrt{3/5}\,Y$ (a convention/model choice, not an algebraic output), equivalently $g_1^2 = \frac{5}{3}g'^2$, the trace ratio at the scale where* SU(5) *is unbroken yields*

$$\sin^2\theta_W = \frac{g'^2}{g^2+g'^2} = \frac{\mathrm{Tr}(T_3^2)}{\mathrm{Tr}(Q^2)} = \frac{1}{1+5/3} = \frac{3}{8}. \tag{9}$$

This is a high-scale boundary value conditioned on the normalization, not the measured low-energy number (Rem. V.2).

Proof. Using $Q = T_3 + Y$ and the normalization condition $\mathrm{Tr}(T_Y^2) = \mathrm{Tr}(T_3^2)$ over a complete SU(5) multiplet gives $\mathrm{Tr}(Q^2) = (1+\frac{5}{3})\,\mathrm{Tr}(T_3^2)$, hence $\sin^2\theta_W = 1/(1+\frac{5}{3}) = 3/8$. Concretely on $\bar{\mathbf{5}} = (d^c : 3 \text{ colors}, Y = +\frac{1}{3}) \oplus (L = (\nu, e), Y = -\frac{1}{2})$,

$$\mathrm{Tr}(T_3^2) = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}, \qquad \mathrm{Tr}(Q^2) = 3\cdot(\tfrac{1}{3})^2 + 0 + (-1)^2 = \tfrac{1}{3} + 1 = \tfrac{4}{3}, \tag{10}$$

so the ratio is $(1/2)/(4/3) = 3/8$. This is Georgi–Quinn–Weinberg [22]. □

Remark V.2 (Honest scale caveat). The value 3/8 is a tree-level boundary condition at M_{GUT}, valid only while SU(5) (or larger) is unbroken; it is *not* the measured number. Renormalization-group running to M_Z yields the experimental

$$\sin^2\theta_W^{\overline{\mathrm{MS}}}(M_Z) = 0.23122 \pm 0.00006 \tag{11}$$

(PDG, $\hat{s}_Z^2$) [23]. The amount of running, and whether the three couplings actually meet, depends entirely on the matter and scalar content between M_Z and M_{GUT}: non-supersymmetric SU(5) does not unify precisely, while the MSSM spectrum does much better. The hypercharge normalization $T_Y = \sqrt{3/5}\,Y$ is the load-bearing input; a different (non-GUT) normalization changes the number.

Remark V.3 (Content-dependence of b_0, and a flagged SJET pitfall). The one-loop gauge beta function is $\beta(g) = -(g^3/16\pi^2)\,b_0$ with

$$b_0 = \tfrac{11}{3}C_2(G) - \tfrac{4}{3}\kappa\sum_F T(R_F) - \tfrac{1}{3}\sum_S T(R_S), \qquad \kappa = \tfrac{1}{2}\ (\text{Weyl}), \tag{12}$$

where $T(\text{fund SU}(N)) = \frac{1}{2}$, $T(\text{adj}) = C_2(G) = N$, $T(\text{singlet}) = 0$ [24, 25]. Every term is a Dynkin index of a *named* representation, so b_0 is content-dependent and is not a universal constant. We explicitly flag the earlier (over-claimed) "$b_0 = 12 = (88-24-28)/3$" construction: a scalar contribution entered as a *coset dimension* (e.g. $\dim E_6/F_4 = 78-52 = 26$) is not a Dynkin index and does not in general equal $\sum_S T(R_S)$. Such a value is therefore a model assumption about field content, not an output of the Lie algebra, and we record it as such rather than as a result.

Remark V.4 (F_4 is non-chiral). F_4 has only real representations, hence cannot produce a chiral spectrum; embedding $\mathrm{SU}(3)\times\mathrm{SU}(2)\times U(1)$ directly into F_4 does not reproduce the correct hypercharges. The chiral content enters through the complexification E_6 and its complex **27**. F_4 enters the program as $\mathrm{Aut}(J_3(\mathbb{O}))$ and E_6 as the reduced structure group of $J_3(\mathbb{O})$; the chiral physics must be routed through $E_6 \to \mathrm{SO}(10) \to \mathrm{SU}(5)$.

VI. THEOREM/CONJECTURE LEDGER

The point of this paper is the transparency of Table II: the structural facts of Secs. II–V are proven (or standard), whereas the dynamical program is conjectural and is stated below only as open problems.

Conjecture VI.1 (Capacity selection of the partition)**.** There exists a well-posed variational or information-theoretic "capacity" principle whose extremization singles out one normed division algebra (or one partition of $J_3(\mathbb{O})$, or one symmetry-breaking pattern of E_6) as physically realized. *Status: open.* No such principle is established here; the heuristic of Sec. I does not constitute one.

Conjecture VI.2 (Observer/family count)**.** The number of fermion generations equals three for a structural reason internal to the octonionic/Jordan data (e.g. via SO(8) triality, three off-diagonal octonionic slots of $J_3(\mathbb{O})$, or the Cl(6) ideal structure). *Status: open and contested.* Triality- and Cl(6)-based generation counting (Furey [29]; Boyle [33]; Todorov–Dubois-Violette [32]) is an active but model-dependent research theme, not a theorem.

Conjecture VI.3 (Mirror-functor generation mechanism)**.** The heuristic "mirror" of Sec. I can be promoted to a precise functor (an involution on a suitable category) whose iteration reproduces the division-algebra chain and generates the matter representations. *Status: open.* As Remark II.5 shows, the only canonical order-two map available—Cayley–Dickson

Proven / standard (this paper, by citation)	**Conjectural / open (not claimed)**
Hurwitz: normed division algebras are $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ (Thm. II.2); sedenion zero divisors (Prop. II.4).	Capacity/selection principle: *why* a particular algebra or partition is dynamically singled out (Conj. VI.1).
$\dim J_3(\mathbb{O}) = 27$; Aut $= F_4$, structure group E_6 (Thms. II.7, II.9).	Observer/family count: *why* three generations (Conj. VI.2).
$H_2(\mathbb{A}) \cong \mathbb{R}^{1,n+1}$, dims $3, 4, 6, 10$; $H_2(\mathbb{H})$ is 6D; 4D is $H_2(\mathbb{C})$ (Thm. III.1, Rem. III.2).	Mirror-functor generation: a precise functor whose action "creates" matter content (Conj. VI.3).
$\mathbf{27} = \mathbf{1} + \mathbf{10} + \mathbf{16}$; $\mathbf{16} = \mathbf{10} + \bar{\mathbf{5}} + \mathbf{1}$; $\mathbf{32} = \mathbf{16} \oplus \overline{\mathbf{16}}$ (Props. IV.1, IV.2, IV.4).	Emergent dynamics: equations of motion / a Lagrangian derived from the algebraic data (Conj. VI.4).
$\mathbf{16} \times \mathbf{16} = \mathbf{10} + \mathbf{120} + \mathbf{126}$ (Prop. IV.5); $\sin^2\theta_W = 3/8$ as a normalization-conditioned high-scale trace ratio (Prop. V.1).	Whether any of the above can yield a falsifiable low-energy prediction beyond the known RG analysis.

TABLE II. The honesty ledger. Left: structural facts, all pre-existing and standard. Right: the dynamical program, recorded as open problems, not results.

conjugation—loses its useful (alternative/triality) structure at $\mathbb{S}$, so even formulating the conjecture precisely is part of the open problem.

Conjecture VI.4 (Emergent dynamics)**.** There is a dynamical theory (action principle and equations of motion) whose kinematic data are exactly the algebraic structures of Secs. II–V and which reproduces Standard-Model dynamics. *Status: open.* This paper supplies only kinematic/representation-theoretic scaffolding; no dynamics is derived, and we make no mass, mixing, or coupling predictions.

VII. CONCLUSION

We have collected, with correct dimensions and signatures and with explicit citations, the standard algebraic facts that connect normed division algebras, the Albert algebra $J_3(\mathbb{O})$, and the exceptional groups F_4 and E_6 to the representation-theoretic skeleton of the Standard Model. The corrections we insisted upon—that the normed chain ends at $\mathbb{O}$ because the sedenions have zero divisors (with conjugation nonetheless remaining an involution), that $H_2(\mathbb{H})$ is six- rather than four-dimensional and that ordinary 4D spacetime is $H_2(\mathbb{C})$, that the EIII tangent space is $\mathbf{16} \oplus \overline{\mathbf{16}}$ and not "$28 + 4$," and that $\sin^2\theta_W = 3/8$ is a high-scale boundary value with content-dependent running—are exactly the places where over-claiming is easy. None of the mathematics is new. The contribution is solely organizational: a single explicitly heuristic motivation, and a transparent ledger that separates what is proven from a dynamical program that remains entirely conjectural. This is a synthesis and an open-problem statement, not a derivation of physics; accordingly it belongs in a foundations / mathematical-physics venue rather than as a claim of new falsifiable prediction.

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[1] J. C. Baez, "The Octonions," Bull. Amer. Math. Soc. (N.S.) **39** (2002), 145–205 (errata **42** (2005), 213). arXiv:math/0105155.

[2] A. Hurwitz, "Über die Komposition der quadratischen Formen," Math. Ann. **88** (1923), 1–25 (posthumous; work dated 1898).

[3] P. Jordan, J. von Neumann, and E. Wigner, "On an algebraic generalization of the quantum mechanical formalism," Ann. of Math. **35**, no. 1 (1934), 29–64.

[4] A. A. Albert, "On a certain algebra of quantum mechanics," Ann. of Math. **35**, no. 1 (1934), 65–73.

[5] C. Chevalley and R. D. Schafer, "The exceptional simple Lie algebras F_4 and E_6," Proc. Natl. Acad. Sci. USA **36**, no. 2 (1950), 137–141.

[6] R. D. Schafer, *An Introduction to Nonassociative Algebras*, Pure and Applied Mathematics 22, Academic Press, New York, 1966 (Dover reprint 1995).

[7] T. A. Springer and F. D. Veldkamp, *Octonions, Jordan Algebras and Exceptional Groups*, Springer Monographs in Mathematics, Springer, Berlin, 2000.

[8] N. Jacobson, *Structure and Representations of Jordan Algebras*, Amer. Math. Soc. Colloquium Publications 39, AMS, Providence, RI, 1968.

[9] J. H. Conway and D. A. Smith, *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*, A K Peters, Natick, MA, 2003.

[10] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, New York, 1991.

[11] G. Moreno, "The zero divisors of the Cayley–Dickson algebras over the real numbers," Bol. Soc. Mat. Mexicana (3) **4** (1998), 13–28.

[12] R. E. Cawagas, "On the structure and zero divisors of the Cayley–Dickson sedenion algebra," Discuss. Math. Gen. Algebra Appl. **24** (2004), 251–265.

[13] J. C. Baez and J. Huerta, "Division Algebras and Supersymmetry I," in *Superstrings, Geometry, Topology, and C*-algebras*, Proc. Sympos. Pure Math. 81, AMS, Providence, 2010, pp. 65–80 (arXiv:0909.0551); "...II," Adv. Theor. Math. Phys. **15** (2011), 1373–1410 (arXiv:1003.3436).

[14] A. Sudbery, "Division algebras, (pseudo)orthogonal groups and spinors," J. Phys. A: Math. Gen. **17**, no. 5 (1984), 939–955.

[15] T. Kugo and P. Townsend, "Supersymmetry and the division algebras," Nucl. Phys. B **221**, no. 2 (1983), 357–380.

[16] J. M. Evans, "Supersymmetric Yang–Mills theories and division algebras," Nucl. Phys. B **298**, no. 1 (1988), 92–108.

[17] T. Dray, J. Janesky, and C. A. Manogue, "Octonionic Hermitian matrices with non-real eigenvalues," Adv. Appl. Clifford Algebras **10** (2000), 193–216. arXiv:math/0006069.

[18] R. Slansky, "Group theory for unified model building," Phys. Rep. **79**, no. 1 (1981), 1–128.

[19] S. Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, Graduate Studies in Mathematics 34, AMS, 2001 (corrected reprint of the 1978 Academic Press edition); classification of irreducible Hermitian symmetric spaces, type EIII.

[20] P. Langacker, "The Physics of Heavy Z' Gauge Bosons," Rev. Mod. Phys. **81** (2009), 1199; arXiv:0801.1345.

[21] H. Georgi and S. L. Glashow, "Unity of All Elementary-Particle Forces," Phys. Rev. Lett. **32** (1974), 438–441.

[22] H. Georgi, H. R. Quinn, and S. Weinberg, "Hierarchy of Interactions in Unified Gauge Theories," Phys. Rev. Lett. **33** (1974), 451–454.

[23] S. Navas *et al.* (Particle Data Group), "Review of Particle Physics," Phys. Rev. D **110**, 030001 (2024); Electroweak Model section, $\hat{s}_Z^2 = \sin^2\theta_W(M_Z) = 0.23122 \pm 0.00006$.

[24] M. E. Machacek and M. T. Vaughn, "Two-Loop Renormalization Group Equations in a General Quantum Field Theory: (I) Wave Function Renormalization," Nucl. Phys. B **222** (1983), 83.

[25] D. R. T. Jones, "Two-loop diagrams in Yang–Mills theory," Nucl. Phys. B **75** (1974), 531.

[26] M. Günaydin and F. Gürsey, "Quark structure and octonions," J. Math. Phys. **14** (1973), 1651–1667.

[27] F. Gürsey, P. Ramond, and P. Sikivie, "A universal gauge theory model based on E_6," Phys. Lett. B **60** (1976), 177–180.

[28] J. C. Baez and J. Huerta, "The Algebra of Grand Unified Theories," Bull. Amer. Math. Soc. (N.S.) **47** (2010), 483–552; arXiv:0904.1556.

[29] C. Furey, "Generations: three prints, in colour," J. High Energy Phys. **2014** (10), 046; arXiv:1405.4601.

[30] C. Furey, "Standard model physics from an algebra?", PhD thesis, University of Waterloo (2015); arXiv:1611.09182.

[31] O. C. Stoica, "Leptons, Quarks, and Gauge from the Complex Clifford Algebra $\mathbb{C}\ell_6$," Adv. Appl. Clifford Algebras **28** (2018), 52; arXiv:1702.04336.

[32] I. Todorov and M. Dubois-Violette, "Deducing the symmetry of the standard model from the automorphism and structure groups of the exceptional Jordan algebra," Int. J. Mod. Phys. A **33** (2018), 1850118; arXiv:1806.09450.

[33] L. Boyle, "The Standard Model, The Exceptional Jordan Algebra, and Triality," arXiv:2006.16265 (2020); published in Quantum Reports **2** (2020).

[34] T. Dray and C. A. Manogue, "Octonionic Cayley spinors and E_6," Comment. Math. Univ. Carolin. **51** (2010), 193–207, and related work on $\mathrm{SL}(2,\mathbb{O})$ and octonionic Dirac equations.

[35] C. A. Manogue, T. Dray, and R. A. Wilson, "Octions: An E_8 description of the Standard Model," J. Math. Phys. **63** (2022), 081703; arXiv:2204.05310.

[36] A. G. Lisi, "An Exceptionally Simple Theory of Everything," arXiv:0711.0770 (2007).

[37] A. G. Lisi, “An Explicit Embedding of Gravity and the Standard Model in E_8,” arXiv:1006.4908 (2010).

[38] J. Distler and S. Garibaldi, “There is no ‘Theory of Everything’ inside E_8,” Comm. Math. Phys. **298** (2010), 419–436; arXiv:0905.2658.